

THE PARITY CONJECTURE FOR INDEPENDENCE POLYNOMIALS OF GENERALIZED PETERSEN GRAPHS IS FALSE

DEEP BHATTACHARJEE

ABSTRACT. It has been conjectured that for $n \geq 2k + 1$ the independence polynomial of the generalized Petersen graph $\text{GP}(n, k)$ has only real roots if and only if k is even. We show that the statement contradicts itself: $\text{GP}(n, 2) \cong \text{GP}(n, (n-1)/2)$ whenever $n \equiv 3 \pmod{4}$, and $(n-1)/2$ is then odd. We also show that each direction fails on its own, at $\text{GP}(7, 3)$ and at $\text{GP}(9, 2)$.

1. INTRODUCTION

For integers $n \geq 3$ and $1 \leq k < n/2$, the generalized Petersen graph $\text{GP}(n, k)$ of Watkins [5] has vertices $u_0, \dots, u_{n-1}$ and $v_0, \dots, v_{n-1}$ and the edges

$$u_i u_{i+1}, \quad u_i v_i, \quad v_i v_{i+k} \quad (0 \leq i < n),$$

with indices read modulo n . The independence polynomial of a graph G is $I(G; x) = \sum_{j \geq 0} a_j x^j$, where a_j is the number of independent sets of G with j vertices; its roots are always real for some families and spread over the plane for others [1]. Real-rootedness of independence polynomials is known for claw-free graphs by the theorem of Chudnovsky and Seymour [2]; generalized Petersen graphs are cubic and contain claws, and their roots were studied numerically by Pandey [3], who proposed the following.

Conjecture 4.1 of [3]. For all integers $n \geq 2k + 1$, the polynomial $I(\text{GP}(n, k); x)$ has only real roots if and only if k is even.

The conjecture assigns a property to the pair (n, k) , whereas real-rootedness depends only on the isomorphism class of $\text{GP}(n, k)$. Different step sizes can give the same graph, and this is enough to refute it.

Theorem 1.1. *The following hold.*

- (a) *If $n \equiv 3 \pmod{4}$ and $n \geq 7$, then $\text{GP}(n, 2) \cong \text{GP}(n, \frac{n-1}{2})$, the step $\frac{n-1}{2}$ is odd, and both pairs $(n, 2)$ and $(n, \frac{n-1}{2})$ satisfy $n \geq 2k + 1$. So Conjecture 4.1 of [3] fails for at least one of these two pairs, for every such n .*
- (b) *$I(\text{GP}(7, 3); x)$ has only real roots, although 3 is odd.*
- (c) *$I(\text{GP}(9, 2); x)$ has exactly five real roots and one pair of non-real roots, although 2 is even.*

Part (a) is a statement about isomorphisms alone and needs no knowledge of the polynomials. Parts (b) and (c) show that neither implication of the conjecture survives separately.

2020 *Mathematics Subject Classification.* Primary 05C31; Secondary 05C60, 26C10.

Key words and phrases. independence polynomial, real roots, generalized Petersen graph, graph isomorphism.

Corresponding author: d.bhattacharjee@erl-forschung.de; ORCID [0000-0003-0466-750X](https://orcid.org/0000-0003-0466-750X).

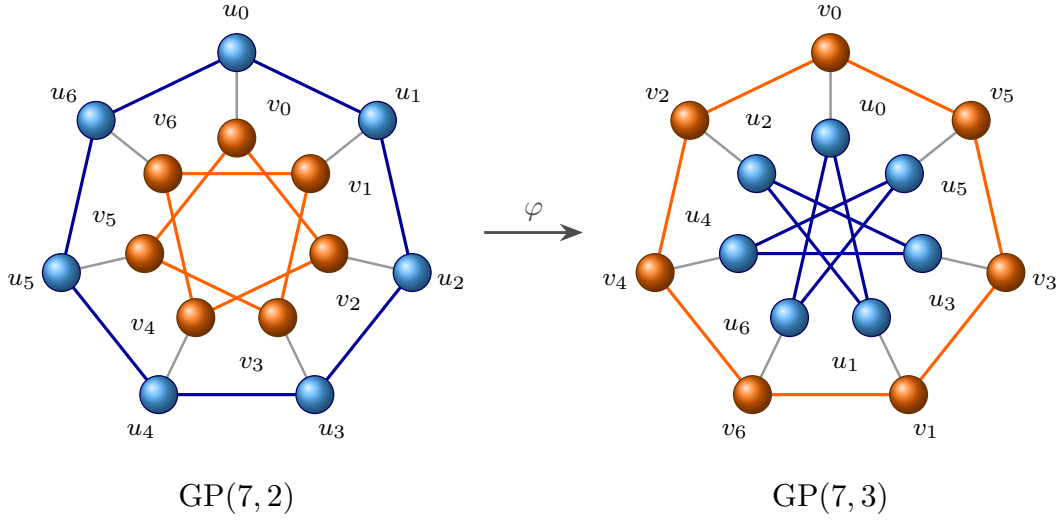


FIGURE 1. The isomorphism φ of Lemma 2.1 for $n = 7$, $k = 2$, $l = 3$. Each vertex of $\text{GP}(7, 3)$ carries the name of its preimage. The outer cycle of $\text{GP}(7, 2)$ (blue) becomes the inner cycle of $\text{GP}(7, 3)$, and the inner cycle (orange) becomes the outer one.

2. ISOMORPHIC GENERALIZED PETERSEN GRAPHS

Lemma 2.1. *Let $1 \leq k, l < n/2$ with $kl \equiv \pm 1 \pmod{n}$. Then the map $\varphi(u_i) = v_{li}$, $\varphi(v_i) = u_{li}$ is an isomorphism from $\text{GP}(n, k)$ onto $\text{GP}(n, l)$.*

Proof. Since $kl \equiv \pm 1$, the number l is invertible modulo n , so $i \mapsto li$ permutes the indices and φ is a bijection. It maps the outer edge $u_i u_{i+1}$ to $v_{li} v_{li+l}$, an inner edge of $\text{GP}(n, l)$; the spoke $u_i v_i$ to the spoke $v_{li} u_{li}$; and the inner edge $v_i v_{i+k}$ to $u_{li} u_{li+lk} = u_{li} u_{li \pm 1}$, an outer edge. Both graphs have $3n$ edges, so φ is an isomorphism. $\square$

Steimle and Staton [4] proved that these maps, together with $\text{GP}(n, k) = \text{GP}(n, n - k)$, account for all isomorphisms between generalized Petersen graphs; we only need the easy direction above.

Proof of Theorem 1.1(a). Let $n \equiv 3 \pmod{4}$, $n \geq 7$, and $l = (n - 1)/2$. Then $2l = n - 1 \equiv -1 \pmod{n}$, so $\text{GP}(n, 2) \cong \text{GP}(n, l)$ by Lemma 2.1. Since $n - 1 \equiv 2 \pmod{4}$, l is odd, and $n = 2l + 1 \geq 2 \cdot 2 + 1$. Conjecture 4.1 of [3] for $(n, 2)$ says that $I(\text{GP}(n, 2); x)$ is real-rooted, and for (n, l) that the same polynomial is not. $\square$

The same argument applies whenever some even k and odd l below $n/2$ satisfy $kl \equiv \pm 1 \pmod{n}$; for instance $4 \cdot 3 \equiv -1 \pmod{13}$ gives $\text{GP}(13, 4) \cong \text{GP}(13, 3)$.

3. THE TWO POLYNOMIALS

Counting independent sets of each size gives

$$(3.1) \quad I(\text{GP}(7, 2); x) = 1 + 14x + 70x^2 + 154x^3 + 147x^4 + 49x^5,$$

$$(3.2) \quad I(\text{GP}(9, 2); x) = 1 + 18x + 126x^2 + 438x^3 + 801x^4 + 747x^5 + 303x^6 + 27x^7.$$

Some coefficients can be read off at once: a_1 is the number of vertices and $a_2 = \binom{2n}{2} - 3n$, since $\text{GP}(n, k)$ has $3n$ edges. Since $2 \cdot 3 \equiv -1 \pmod{7}$, Lemma 2.1 gives $\text{GP}(7, 3) \cong \text{GP}(7, 2)$, so (3.1) is also the independence polynomial of $\text{GP}(7, 3)$ (Figure 1).

Proof of Theorem 1.1(b). Let P be the polynomial (3.1). Its values at seven points are

x	$-\frac{7}{5}$	$-\frac{13}{10}$	-1	$-\frac{3}{5}$	$-\frac{3}{10}$	$-\frac{1}{5}$	$-\frac{1}{10}$
$P(x)$	$-\frac{8733}{3125}$	$\frac{67513}{10^5}$	1	$-\frac{697}{3125}$	$\frac{1363}{10^5}$	$-\frac{39}{3125}$	$\frac{16021}{10^5}$

The sign changes five times along the table, so P has a root in each of the five disjoint intervals $(-\frac{7}{5}, -\frac{13}{10})$, $(-1, -\frac{3}{5})$, $(-\frac{3}{5}, -\frac{3}{10})$, $(-\frac{3}{10}, -\frac{1}{5})$ and $(-\frac{1}{5}, -\frac{1}{10})$. As P has degree five, all its roots are real. By the remark above, $P = I(\text{GP}(7, 3); x)$, and 3 is odd. $\square$

Proof of Theorem 1.1(c). Let Q be the polynomial (3.2). Its sign at the ten points below changes across each of the five intervals

$$\begin{aligned} Q(-8.28862) < 0 < Q(-8.28861), & \quad Q(-0.51657) > 0 > Q(-0.51656), \\ Q(-0.31262) < 0 < Q(-0.31261), & \quad Q(-0.22263) > 0 > Q(-0.22262), \\ Q(-0.16871) < 0 < Q(-0.16870), \end{aligned}$$

so Q has real roots $\rho_1, \dots, \rho_5$, one in each interval. Let r and r' be the two remaining roots, counted with multiplicity. Since Q has degree seven with leading coefficient 27, the relations between roots and coefficients give

$$r + r' = -\frac{303}{27} - \sum_{i=1}^5 \rho_i, \quad rr' = -\frac{1}{27 \rho_1 \rho_2 \rho_3 \rho_4 \rho_5}.$$

Both numbers are real. Adding the endpoints of the intervals gives $-9.50915 < \sum \rho_i < -9.50910$, so $-1.713123 < r + r' < -1.713071$ and $(r + r')^2 < 2.9348$. All ρ_i are negative and $|\rho_1 \cdots \rho_5| < 8.28862 \cdot 0.51657 \cdot 0.31262 \cdot 0.22263 \cdot 0.16871 < 0.050278$, so $rr' > 1/(27 \cdot 0.050278) > 0.7366$ and $4rr' > 2.9464$. Hence

$$(r - r')^2 = (r + r')^2 - 4rr' < 0,$$

so $r - r'$ is not real, and r and r' are complex conjugates that are not real. Thus $Q = I(\text{GP}(9, 2); x)$ has exactly five real roots, although 2 is even. $\square$

Remark 3.1. The non-real pair of $I(\text{GP}(9, 2); x)$ is close to $-0.85655 \pm 0.05554i$. In [3] the polynomials were computed with a transfer matrix whose entries carry the weight $x^{|B|}$, where $|B|$ is the number of occupied positions in a state that also records earlier inner vertices; with this weighting a vertex is counted once for every state in which it appears. The report there of 50 roots for $\text{GP}(25, 2)$ reflects this: the spokes $u_i v_i$ cover every vertex and an independent set meets each spoke at most once, so $I(\text{GP}(25, 2); x)$ has degree at most 25. The polynomials behind the numerical evidence in [3] are therefore not independence polynomials.

Use of generative AI. The author used Claude (Anthropic) for literature search, computation and drafting; the author checked the mathematics and takes full responsibility for the content.

REFERENCES

- [1] J. I. Brown, C. A. Hickman and R. J. Nowakowski, *On the location of roots of independence polynomials*, J. Algebraic Combin. **19** (2004), 273–282. <https://doi.org/10.1023/B:JACO.0000030703.39946.70>
- [2] M. Chudnovsky and P. Seymour, *The roots of the independence polynomial of a clawfree graph*, J. Combin. Theory Ser. B **97** (2007), 350–357. <https://doi.org/10.1016/j.jctb.2006.06.001>
- [3] R. Pandey, *Parity-dependent real-rootedness in independence polynomials of generalized Petersen graphs*, preprint, 2026. [arXiv:2601.03293](https://arxiv.org/abs/2601.03293)
- [4] A. Steimle and W. Staton, *The isomorphism classes of the generalized Petersen graphs*, Discrete Math. **309** (2009), 231–237. <https://doi.org/10.1016/j.disc.2007.12.074>
- [5] M. E. Watkins, *A theorem on Tait colorings with an application to the generalized Petersen graphs*, J. Combin. Theory **6** (1969), 152–164. [https://doi.org/10.1016/S0021-9800\(69\)80116-X](https://doi.org/10.1016/S0021-9800(69)80116-X)

FORMERLY, ELECTRO-GRAVITATIONAL SPACE PROPULSION LABORATORY (EGSPL),
BHUBANESWAR, ODISHA 751030, INDIA

Email address: d.bhattacharjee@erl-forschung.de

Email address: itsdeep@live.com

URL: <https://orcid.org/0000-0003-0466-750X>